The statement of the theory

Three Claude Fable discussants in four rounds, 29 and 30 October 2026

The three discussants took the learning attacker, the defence and the steering attacker. Each stated your idea on its aspect, critiqued it and formulated it again, and the three then agreed on one statement of the theory in 26 clauses. The statement is sharper than what we had, and it defines the three objects the theory lacked: a declared learner, a defence and a steering attacker. Its central finding is that no lower bound sees an attacker because it learns, since fine-precision SGD simulates every sample-based learner (AKMSS21 Theorem 1a). A bound sees a learning attacker through what it reads, through the noise of its own updates, or through an assumption on every efficient algorithm. The one new tool is the second handle applied to the attacker. Three of your ideas do not survive as stated, and section 2 says which. Each discussant ended with one statement to prove and one experiment for the oracle. All three plant nothing, none gives hardness, and the core stays open. Nothing here enters the booklet or the registers before you rule. Files are in `lab/theory/council/2026-10-29-statement-of-the-theory/`, the statement in `statement/final.md`.

1. The round

Your ideas from the meeting entered as the integrator's paraphrase in `halil-ideas.md`, with nothing quoted and nothing of Küpçü's, ranked below `CONTEXT.md` and the established results. In round 00 all three discussants ranked the learning attacker first. By the rule of the round it went to the earliest reply, and the others took their next free choice. Your capacity idea was ranked third by two discussants and folded into the learning attacker. In round 01 each wrote on its aspect, in round 02 each judged the other two's objects (22 verdicts, 12 accepted, 10 revised, none rejected, 9 blocking challenges), and in round 03 the learning-attacker discussant edited one statement, which the other two reviewed clause by clause (19 accepted, 23 amended, none rejected). The editor took 21 of the 23 amendments and recorded 2 dissents with reasons. In round 04 each thought toward your idea under the final statement.

2. What the discussants said about your ideas

The learning attacker pays what the trainer paid (your item 1). False in both directions for a general secret, and exact under two conditions together: the attacker's observations are the trainer's data in law, and its secret is the trainer's own target as a learning problem (clause 12, proven in the Gaussian linear model and for correlational classes). Each condition is needed. A proper function of the target can cost d times less, and an exact release is relearned from d queries.

A learning-constrained attacker meets a sharper bound (item 1). The word learner restricts nothing, by AKMSS21 Theorem 1a. A bound has three handles, listed above, and the theory had used the noise handle on the trainer alone (item 18 of the reading of 29 October). For a backdoor or a watermark the

conjecture attaches to the wrong side. The attacker knows its trigger or key and learns nothing, so the learner is the auditor or the user (CHLX25 Definition 3.1 and Theorem 4.1, CGZ24 Lemma 4, where a watermark's price is its key length).

Training is post-processing, so planting before it adds nothing (item 9). Right for a distinguishing auditor who holds the data, by data processing. It fails three ways. An auditor of the model alone faces the trainer's filter, which is a task made hard by learning and already in the theory (item 18). For an attacker that steers, training is the mechanism and not post-processing. The published undetectable backdoors plant in the training coins or in a frozen layer, with no training in between (GKVZ22 Theorem 6.4, BRV26 Theorems 2 and 5), so your word for them, ugly, holds for the sharper reason that they say nothing about learning.

The planting objection (item 4). It does not reach structure the attacker plants, since the attacker's choices are moves of the game. That structure yields attack results only, priced as the auditor's detection price, and never a defence theorem.

A designer's modification indistinguishable from an honest sample (`CONTEXT.md`, 27 October). The right rule for measurement, and it admits no defence, since an admissible programming moves no price (item 10). A canary that replaces an honest record programs and measures. Release noise changes the honest procedure and defends. Your decision 5 is settled this way if you accept it.

A constraint honest parties already meet (item 10). It describes practice and does no mathematical work. Clause 5 replaces it. A defence draws all its randomness fresh from a public law at each application, and no other node, party or later application reads it. It runs at every execution, and keeps the trainer's model family and loss. Its lower bound holds over an instance class fixed before the defence. So the planting ban lives in the instance class. A random layer reused across the records of one run is a kept key and fails.

Capacity as the price (item 11). Agreement on held-out data is a success level, not a budget, and the parameter count is a coordinate of the budget, not a price (clause 3). Against the noisy gradient attacker, capacity lowers the price by at most the factor p (clause 13).

White box (item 12). The white-box attacker is bound only by statements that bind every observer or that restrict the trainer (clause 20). The linear readout class practitioners run has the degree-two price, within a constant of every observer on generic laws (DPLB24 Proposition 2.6).

3. The statement, in brief

The full text is `statement/final.md`. These clauses carry its content.

- Objective. Price an adversary in the units of the procedure's training, prove the price by reduction, measure it with the oracle. The headline target is a natural procedure where a declared attacker's price exceeds the trainer's by a named and measured margin. Item 15 of 29 October is the one instance.
- Clause 3, the declared learner. A model family of p parameters, a loss, an optimizer with batch b, steps T and precision or injected noise, and a readout fixed in advance. The trainer is one. The attacker and auditor are declared learners when a statement says so, and otherwise members of a declared class. A learning attacker's price exists only when the hidden frame is drawn per instance with positive entropy, since on a fixed instance a fixed readout may be the Bayes decoder.
- Clause 5, the defence, as in section 2, with a utility price paid by each honest party in its own coordinates and a security price as a ratio of the attacker class's prices.

- Clause 7, the steering attacker. Setting records, sending an update and altering a value in transit are one object, a shift $v_t(\theta)$ on the trainer's update stream, with a deviation budget and a push-to-noise budget. A chosen seed is a shift of zero, priced separately by clause 22.
- Clause 10, one lemma with three uses. Under two laws the trajectory laws of a Gaussian update chain differ in total variation by at most $\sqrt{T\bar{\kappa}/(4\varsigma^2)}$. It bounds the trainer as a filter, the noisy gradient attacker in steps, and the steerer's stealth.
- Clause 13, the noisy gradient attacker (proven up to degree three, heuristic above). Its price in steps is of order $\varsigma^2 d^{\lceil (l+1)/2 \rceil}/G_{\mathrm{V}}^2$, with l the information exponent and G_{V}^2 the family's squared gradient norm. The eigenvalues behind it are $1/d$, $2/(d(d+2))$ and $3/(d(d+2))$ at degrees one to three, computed by the discussants and not checked by the integrator.
- Clause 14, the stealth cap (proven in the linear model for a known direction). A stealthy label poisoner of noisy gradient descent moves the trigger response by at most $m(\delta) = 2\Phi^{-1}((1+\delta)/2)$ honest standard deviations, and an auditor of the trajectory gains the factor $\sqrt{\eta\mu T/2}$.
- Clause 15. An auditor that holds the trainer's own statistics at the trainer's tolerance detects every steer, so stealth against it is impossible wherever steering happens.
- Clauses 16, 24 and 25, the defences. Gaussian releases, noisy gradients and normalization bind every observer. No bound on the trainer's output alone filters. Local randomizers defend by composition, since a trainer reading records through them is a statistical-query learner (Kasiviswanathan and coauthors, FOCS 2008, as registered).
- Clause 21. The oracle supplies baselines by drawn runs, the response and leakage certificates by coupled runs, and both prices of a defence. A measured failure of a declared attack is never a lower bound.
- Clauses 18 and 26, open. A bound for a batch-one gradient attacker at fine precision without injected noise outside the single-index family, the core. And a rule for when a price measured on real data, with no stated law, meets a proven side on a built law. Until that rule exists a result on real data is the measured price of a declared attack and no more.

4. The three statements to prove

`thm-batching-removes-the-class` (learning attacker, sketch). In the update stream at the square-loss optimum, on a generic law with $c_2 \neq 0$, the ratio of the linear attacker's price to every observer's tends to the spiked-covariance ratio as the batch b grows, which tends to one in the dimension. At finite b the excess is of order $\mathrm{Var}(W^2 T(W))/(b\, c_2^2)$. So the attacker's class enters the price only through the non-Gaussian part of the update, which batching removes. It plants nothing and is a claim about a mechanism of training. Its spiked-covariance step cites Perry, Wein, Bandeira and Moitra (Annals of Statistics 2018) and Onatski, Moreira and Hallin (Annals of Statistics 2013) from memory. The step expected to fail is the order of limits, since the per-update error compounds over about d updates and needs $b \gg \sqrt{d}$. Experiment: $d \in \{200, 400\}$, $b \in \{1, 4, 16, 64\}$, labels $W = \min(|Z|, 4)$, PCA against the norm-weighted estimator, 40 runs per cell. A ratio above the predicted band at $b = 64$ refutes it.

`thm-noise-exponent-gap` (defence, sketch). With honest Gaussian noise on client updates and $x = b\varsigma^2/\mu_2$, every observer's price is $\Theta(d(1+x)^2)$ and the server's training price is $\Theta(d(1+x))$. So the attacker pays $\Theta(1+x)$ times the trainer at matched budgets, and the noise exponent of each party's price equals the degree of its channel. The proof conditions on the label magnitudes and uses the Mehler contraction of the Ornstein and Uhlenbeck semigroup, then Fano and PCA. The step expected to fail is a uniform bound on the conditional chi-square at the reference noise. If it fails, the exponent drops to one and the theorem is lost. Its consequence is a measuring device, the degree of a channel on real data read as

a slope against honest noise (heuristic beyond Gaussian channels). Experiment: fit the slopes of PCA's price and the server's rounds against $1 + x$ at $x \in \{1, 4, 16, 64\}$. Slopes outside 2 ± 0.15 and 1 ± 0.15 refute it.

`thm-stealth-uncertainty-cap-linearized` (steering, open). For every model linear in its parameters with fixed features φ, a linear model, random features or a network in its linearized regime, trained by noisy gradient descent, a steer stealthy at level δ against every auditor of the final model moves the prediction at any input x by at most $m(\delta)$ times the honest standard deviation of $f(x)$ over seeds. One shift along the trigger's features attains the cap. An attacker without the features reaches a fraction $1 - C\sqrt{p_{\text{eff}}/N}$ of it from N shadow runs, and at a matched noise the honest law is the Bayesian posterior. The cap is in the trainer's units and shrinks where the trainer has learned, so a stealthy backdoor can live only where the trainer is uncertain. The step expected to fail is the shadow-run rate at triggers outside the span of the training features, the part a backdoor exploits. Beyond the theorem, the transfer to feature learning is open. Experiment: MNIST 3 against 8, two-layer ReLU at widths 64 and 4096 beside the frozen-feature control, label poisoning aimed along $\varphi(\tau)$, 200 runs per side. A ratio above one on the control refutes the proof, and above one at width 4096 refutes the transfer.

5. What this means

The statement is better than what the theory had. It defines the learning attacker, which the definitions reading of 27 October named as the missing object, and it puts defences and steering into the price. It also locates your idea precisely. The class of an attacker enters a price through what it reads, and on a generic law the linear attacker differs from every observer by a constant. An exponent gap between a gradient learner and every observer is known only through the information exponent, whose low moments vanish, and that is the knife edge the council rejected (round 04 of the learning-attacker discussant, heuristic as a reading of the register).

Of the three statements, the stealth cap is closest to your sentence that a task should become hard because of learning. It bounds an acting attacker by what the trainer has not yet learned, in the trainer's units, measured by drawn runs. It is an attack bound in a linearized model, and feature learning is where it is expected to fail. The noise-exponent gap is the second instance of the same-units gap, constant in d and growing with the noise. Batching is a bound in constants. None reaches the core. Several citations in section 4 are from memory and need the search gate before any object enters the booklet.

Four decisions are yours.

1. Should the definitions of clauses 3, 5 and 7 (declared learner, defence, steering attacker) become the theory's definitions? They also settle decision 5 of the reading of 29 October, the narrow reading of the programming rule.
2. Do you accept capacity as a coordinate of the budget and not a price, or do you want it argued again?
3. Which statement first: the stealth cap, the noise-exponent gap, or batching?
4. Should a round go to the open rule of clause 26, when a measured price on real data counts as meeting a proven one? Every result on real models depends on that rule.